Keeping User-Generated Text In-House: Reducing Third-Party Review Processing for Frozen Sequential Recommenders

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Abstract

Hosted text encoders can expand the external processing boundary of recommender systems by transferring user-generated reviews to third parties. Self-hosted neural encoders avoid that transfer but add GPU dependencies, while many text-aware recommenders require retraining. We study whether review signals can improve a deployed sequential recommender while keeping review representation local and the backbone frozen. We propose IHR, a post-hoc reranker that combines frozen backbone scores with locally constructed review profiles. Across six sequential backbones and five Amazon domains, IHR improves all 30 backbone–domain combinations. Strong IHR-augmented systems also outperform retraining-based text-aware baselines under a common evaluation protocol. A controlled representation study shows that CPU-only TF-IDF with SVD outperforms five dense encoders, including Qwen3-Embedding-4B and OpenAI text-embedding-3-large. The gain is driven by rare, item-discriminative terms that complement ID-based collaborative signals. IHR therefore reduces additional third-party review processing without an observed utility penalty in these benchmarks. This is an operational data-governance result, not a formal privacy guarantee: high-IDF identifiers can themselves create linkability risk and should be audited.

CCS Concepts

• **Information systems** → **Recommender systems**; *Retrieval models and ranking*; • **Security and privacy** → **Privacy protections**.

Keywords

Sequential Recommendation, Data Governance, Third-Party Processing, Local Reranking

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1 Introduction

User-generated text can improve recommendation, but incorporating it can expand the data-processing boundary of an AI service. Hosted embedding APIs require transferring review text to a third party, whereas self-hosted neural encoders add local inference infrastructure. Text-enhanced recommenders also commonly retrain the recommendation model around the new representation, requiring another pass over historical interaction data. These dependencies matter when an operator wishes to limit external processing, preserve data residency, or modify an already deployed recommender as little as possible. We therefore ask: *can review text improve a deployed sequential recommender while keeping review representation local and the backbone frozen?*

Sequential recommenders such as GRU4Rec [1], SASRec [3], BERT4Rec [4], FMLPRec [5], DuoRec [6], and BSARec [7] learn from interaction sequences but do not directly use the preference reasons and item-specific descriptors present in reviews. Representative text-aware methods instead modify or retrain the backbone to incorporate side information [9, 10, 13, 15]. Existing reranking work primarily targets diversity, calibration, or learned candidate interactions rather than adding a new review-derived relevance signal to an unchanged sequential backbone [21, 24]. This leaves a deployment gap for systems whose backbone is already trained and whose review corpus should remain within operator-controlled infrastructure.

We address this gap with IHR (in-house reranking), a post-hoc reranker that combines frozen backbone scores with locally constructed review profiles. The default profile uses TF-IDF with truncated SVD and requires neither a hosted text encoder nor backbone retraining. Across six backbones and five Amazon domains, IHR improves all 30 backbone–domain combinations. Because profile construction is decoupled from the reranking interface, we also compare alternative representations under the same backbone and evaluation setup. Local TF-IDF outperforms all five dense encoders we test, including Qwen3-Embedding-4B and OpenAI text-embedding-3-large.

The result has a data-governance interpretation but not a formal privacy one. The highest-utility configuration in our experiments also avoids transferring the review corpus to a hosted representation service and avoids retraining the deployed recommender. We do not claim differential privacy, resistance to inference attacks, or confidentiality against an adversary. Indeed, the gain is driven by rare item identifiers whose sparsity makes them auditable but can also increase linkability.

Our contributions are threefold. (i) We introduce a post-hoc review reranker for frozen sequential backbones and show consistent gains across all 30 backbone–domain combinations. (ii) We provide a controlled representation comparison showing that locally computed TF-IDF with SVD outperforms substantially larger neural encoders because item-discriminative signal complements the collaborative backbone. (iii) We characterize the operational deployment boundary through top-K efficiency, review-scarcity and hyperparameter robustness, computational cost, and an explicit distinction between reduced third-party processing and formal privacy protection.

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2 Related Work

Sequential and text-enhanced recommendation. Sequential recommendation has evolved from recurrent and convolutional models such as GRU4Rec [1] and Caser [2] to attention-, frequency-, and contrastive-based models including SASRec [3], BERT4Rec [4], FMLPRec [5], DuoRec [6], FEARec [8], and BSARec [7]. These models primarily learn from item-ID interaction sequences. Text- and feature-aware approaches instead incorporate auxiliary information during training, including FDSA [9], S3-Rec [12], DIF-SR [10], UniSRec [13], VQ-Rec [14], TedRec [11], and Recformer [15]. Review-based recommenders likewise encode user or item reviews using neural architectures [16–20]. Unlike these methods, IHR adds review-derived relevance after training while leaving the sequential backbone frozen.

Post-hoc reranking and complementary representations. Classical and learned reranking methods have primarily targeted diversity, calibration, or candidate interactions, including MMR [22], DPP-based reranking [23], PRM [21], and LeapRec [24]. Training-free text recommendation has also been studied in cold-start settings where text directly determines ranking [25]; our setting instead uses review information to complement an already trained collaborative backbone. IHR further draws on classical sparse and distributed representations, including TF-IDF and latent semantic analysis [26, 27], Word2Vec and Doc2Vec [28, 29], collaborative embeddings such as Item2Vec and Prod2Vec [30, 31], and contextual encoders such as SBERT and SimCSE [32, 33]. Prior ensemble and hybrid-recommender work suggests that non-redundant signals can be more useful in combination than individually strong but overlapping ones [35, 36], motivating our complementarity analysis.

Operational data processing and privacy. Recommendation privacy research has shown that sparse behavioral data and recommender outputs can expose sensitive information [37, 38], while differential privacy and federated learning address formal privacy and decentralized-training objectives [39, 40]. Our concern is narrower: whether review representation requires third-party processing and whether introducing text forces retraining of an already deployed recommender. IHR can avoid these two additional dependencies when profiles are constructed locally, but it does not provide protection against re-identification, inference attacks, or disclosure of exported profiles.

3 Method: IHR

IHR is a post-hoc reranker that sits on top of a frozen sequential backbone and integrates a review-based content signal by score-level late fusion. The backbone can be any sequential model that produces per-item scores, and none of its parameters are modified. For user u and candidate item j, the final score is defined as

$$s_j^{\text{final}} = (1-\alpha)\,\hat{s}_j^{\text{bb}} + \alpha\,\hat{s}_j^{\text{content}}, \quad \alpha \in [0,1], \tag{1}$$

where $\hat{s}_j^{\text{bb}}$ is the (normalized) backbone score, $\hat{s}_j^{\text{content}}$ is the (normalized) review-based content score, and α is the blending coefficient that controls the contribution of each signal. At α=0 the system reduces to the backbone alone, and larger α gives more weight to review evidence. The reranking layer requires no backbone or neural reranker training and can be run on CPU; the TF-IDF/SVD profile is fitted once as an unsupervised preprocessing step.

A key property of this design is that representation is decoupled from training. The content score is computed from per-item profile vectors $\hat{\mathbf{p}}_j \in \mathbb{R}^d$, and the profile construction is an independent module. Any procedure that maps each item to a fixed-dimensional vector, whether based on TF-IDF, dense sentence encoders, learned collaborative embeddings, or something else, can be plugged into the same reranking interface. This decoupling has two consequences. It makes IHR a drop-in layer on an already deployed backbone, and it also lets us compare different profile constructions under a single fixed framework (Section 5.2), which we use later in the paper. Figure 1 shows the overall structure.

For the rest of this section we describe the default instantiation that we adopt throughout the paper, a TF-IDF with SVD profile, and defer the comparison against other choices to the experiments.

3.1 Operational exposure boundary

We use *reduced external processing* in a narrow deployment sense. Relative to a frozen sequential backbone, we treat two additional processing steps as avoidable dependencies: (1) transferring raw review text outside operator-controlled infrastructure to obtain embeddings from a hosted service, and (2) retraining the sequential backbone on the newly introduced text representation, which requires reprocessing historical recommendation data and produces a new model fitted to that data. IHR can avoid both when its item profiles are constructed locally. This boundary is deliberately narrower than a privacy threat model: we do not claim protection against compromise of the operator, inference from the backbone, membership inference, or re-identification from shared profiles. The security/privacy relevance is therefore reduced data movement and reduced dependence on external processing, not a formal confidentiality guarantee.

3.2 Stage 1. Building item profiles (offline, once)

For each item j, we aggregate all of its review texts into a single document truncated at length $L_{\max}$ so that items with extremely long reviews do not dominate,

$$d_j = \text{truncate}(\text{concat}(r_{j,1}, \ldots, r_{j,|R_j|}),\ L_{\max}), \tag{2}$$

and items with no review become zero vectors.

Leak prevention. Profiles are built separately for validation and test phases. At validation time, reviews that users left on their own test or validation items are both excluded. At test time, only reviews on the test item itself are excluded, since validation items are already observed during backbone training under the leave-last-out split. This keeps the split strict at profile construction time without artificially hiding information that the backbone has already seen.

We apply TF-IDF vectorization to the item document corpus to obtain $\mathbf{x}_j \in \mathbb{R}^V$, using scikit-learn's smooth IDF,

$$\text{IDF}(w) = \log \frac{1+N}{1+\text{df}(w)} + 1, \tag{3}$$

which assigns high weight to terms appearing in few items and suppresses terms appearing in almost every item so that item-level differences are captured. We then apply Truncated SVD to reduce

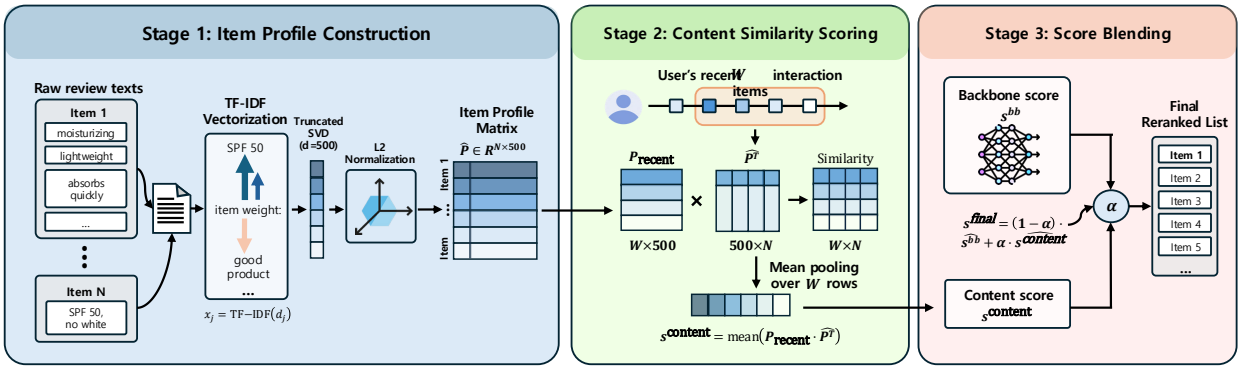


Figure 1: The IHR pipeline. Stage 1 is performed once offline, while Stages 2 and 3 are performed per user at inference time. The sequential backbone remains frozen and no neural reranker is trained.

$\mathbf{x}_j$ to d dimensions and L2-normalize,

$$\hat{\mathbf{p}}_j = \mathrm{SVD}_d(\mathbf{x}_j) \,/\, \|\mathrm{SVD}_d(\mathbf{x}_j)\|_2, \tag{4}$$

producing the item profile matrix $\hat{\mathbf{P}} \in \mathbb{R}^{N \times d}$.

3.3 Stage 2. Content similarity (online)

For user u, let $\mathcal{R}$ be the set of the user's last W items that have valid profiles. The content score for candidate j is the mean similarity,

$$s_j^{\text{content}} = \frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} \hat{\mathbf{p}}_i^\top \hat{\mathbf{p}}_j, \tag{5}$$

computed as a single matrix multiplication $\mathbf{P}_{\text{recent}} \cdot \mathbf{P}^\top$ followed by column-wise mean pooling.

3.4 Stage 3. Score blending

We apply per-user min-max normalization to both s_j^{bb} and s_j^{content} over the unmasked candidate set C (items already seen by the user and padding are masked to -10^9), and then combine them using Eq. (1). The choice of α is described in the experimental setup.

3.5 Computational complexity

Offline, TF-IDF takes $O(N \cdot \bar{L})$ and SVD takes $O(N \cdot V \cdot d)$ and is run once. Online, reranking costs $O(W \cdot N \cdot d)$ per user. No backbone retraining, GPU, neural reranker training, or external API is required for the default local profile configuration.

4 Experimental Setup

This work answers three research questions. RQ1: does IHR consistently improve frozen sequential backbones without retraining them? RQ2: can a locally computed review representation reduce third-party text processing without sacrificing recommendation utility, and what signal property explains the outcome? RQ3: is the method practical and robust to hyperparameters, top-K reranking, and review availability?[1]

[1]Code and data are available at https://anonymous.4open.science/r/TERRA-D8CC/.

Generative AI use in the research workflow. The research question, hypotheses, and experimental design were developed by the authors. After the experimental plan was fixed, generative AI tools were used to assist with implementing the author-specified experimental procedures, executing those procedures, organizing experimental outputs, and drafting and revising the manuscript. The authors independently reviewed the experiment configurations, numerical outputs, statistical analyses, citations, and interpretations reported in this paper and take full responsibility for the work.

4.1 Datasets and protocol

We use five Amazon 2014 5-core datasets [34], where each user and each item has at least five interactions. Table 1 summarizes the statistics. We follow the leave-last-out protocol (last interaction for test, second-last for validation, rest for training) with full-ranking evaluation over all items. The blending coefficient α is swept over $\{0.0, 0.05, 0.10, 0.15, 0.20, 0.30\}$, the value that maximizes validation NDCG@10 is selected, and the corresponding test NDCG@10 under that fixed α is reported, with validation and test item profiles built separately under the leak-prevention rule from Section 3. Truncation length is $L_{\max}$=5,000, and TF-IDF uses V=10,000, minimum document frequency 3, maximum document frequency ratio 0.3, and English stop-word removal.

Table 1: Dataset statistics.

Dataset	Users	Items	Reviews
Beauty	22,363	12,101	198,502
Toys & Games	19,412	11,924	167,597
Sports & Outdoors	35,598	18,357	296,337
Clothing, Shoes & Jewelry	39,387	23,033	278,677
Video Games	24,303	10,672	231,780

4.2 Backbone models

We evaluate six sequential backbones that span different architectures and strength levels: GRU4Rec [1], SASRec [3], FMLPRec [5],

BERT4Rec [4], DuoRec [6], and BSARec [7] (with c=5, α_{BSA}=0.7). Common settings are hidden size 64, two layers, maximum sequence length 50, batch size 256, dropout 0.5, early stopping patience 10, and up to 200 epochs. The learning rate is 0.0005 for BSARec and 0.001 elsewhere. All backbones are trained with three random seeds (42, 123, 456).

4.3 Comparison methods

For a fair comparison, only item-profile construction varies within the interpolation framework of Eq. 1. All profile methods use the same backbone scores, W=5, mean aggregation, evaluation protocol, and validation selection of $\alpha \in \{0, .05, .10, .15, .20, .30\}$ independently per backbone and seed. Thus, within this comparison, only the profile source changes. LLM recommenders requiring multi-billion-parameter inference are outside this computational regime and are not included.

(1) RandomRank. A random unit vector (d=500). A control for the framework itself.
(2) Item2VecRank [30]. A skip-gram embedding (d=500, window 5, 30 epochs) trained on interaction sequences. Chosen as a controlled profile that is high-quality (widely validated in CF) and at the same time of the same kind as the backbone (learned from item co-occurrence), so that it tests the pattern's prior prediction (Section 5.2).
(3) Doc2VecRank [29]. Doc2Vec embedding of each item's reviews (d=256, window 5, 30 epochs).
(4) ReviewEmbRank. Mean-pooled Qwen3-Emb-0.6B review embeddings (d=1024).
(5) IHR (ours). TF-IDF with SVD (d=500); no neural encoder or recommender retraining.

4.4 Metrics and hyperparameter policy

We report Recall@k and NDCG@k for $k \in \{5, 10, 20\}$, with NDCG@10 as the primary metric, averaged over three seeds. Dimensions are 256 for Doc2Vec, 1024 for Qwen3-Emb, and validation-best d=500 for RandomRank, Item2VecRank, and IHR. Figure 3 additionally shows that IHR still improves a strong backbone by +2.7% at d=50.

5 Results and Analysis

5.1 RQ1. Does IHR consistently improve various sequential backbones?

We apply IHR to six backbones across five datasets and report the mean and standard deviation over three seeds (42, 123, 456). Table 2 shows the NDCG@10 results.

Every backbone-dataset combination shows positive improvement, indicating that the observed gain is not confined to a particular backbone, dataset, or initialization. The magnitude has a clear inverse relationship with backbone strength. The weak backbone GRU4Rec improves most (Beauty +94.1%, Toys +266.3%, Clothing +194.9%) and the strong backbone BSARec least (Beauty +4.3%, Toys +11.5%, Sports +6.0%), consistent with a complementarity ceiling in which a stronger backbone leaves less residual room for an auxiliary content signal. The best α is likewise consistent: 0.20 to 0.30 for weak backbones and 0.05 to 0.10 for strong ones (Figure 2). Standard deviations are small in most settings (e.g. Beauty BSARec .0618 ± .0010).

Head-to-head with retraining-based text-aware methods. Table 3 compares complete recommendation systems with end-to-end models that integrate text or item attributes during training (TedRec [11], UniSRec [13], DIF-SR [10], FDSA [9]). BSARec + IHR and DuoRec + IHR achieve the highest reported Recall@10 and NDCG@10 across the five domains under our common protocol. Because these systems use different backbone architectures, this is a system-level comparison rather than an isolated estimate of the causal effect of post-hoc reranking.

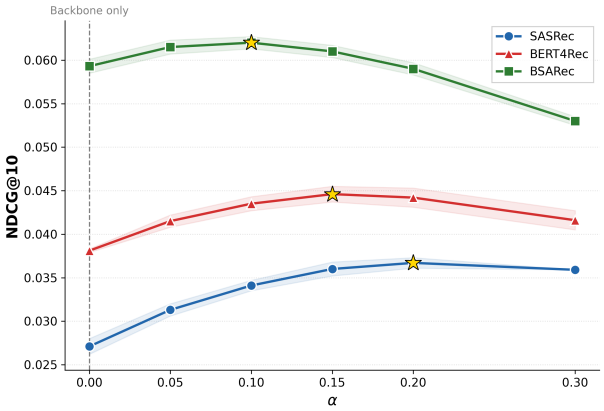


Figure 2: NDCG@10 as a function of the blending coefficient α (Beauty, three representative backbones). The strong backbone BSARec peaks near α=0.10, the weak backbone SASRec near α=0.20. The star marks the best α for each backbone.

To verify complementarity directly, we compare hit items between backbone only (R@10) and backbone with IHR (Table 4). For SASRec, 45.1% of the combined hits are items the backbone would have missed. The ratio shrinks with backbone strength (BSARec 12.6%), but the method still adds 292 new hits for a net gain of 115 items.

5.2 RQ2. Can local review representation reduce third-party processing without utility loss?

We compare five profile sources within the same interpolation framework (same backbone score, α sweep, and evaluation protocol), isolating the effect of profile construction. (Table 5)

The local TF-IDF profile outperforms the neural profile alternatives in this controlled comparison. A three-step argument explains this using the redundancy-complementarity pattern we observe.

Step 1. The framework is neutral (RandomRank). RandomRank yields 0.0% on BERT4Rec and BSARec (where α=0 is always selected) and −0.6% on SASRec (where some seeds pick a small α>0 whose random profile slightly hurts at test time), so any improvement must come from the profile itself.

Step 2. Prior prediction and its test (Item2VecRank). A natural hypothesis is that a higher-quality profile yields larger improvement (the quality hypothesis). A redundancy-complementarity perspective, drawn from the ensemble diversity literature [35, 36]

Table 2: Main results for IHR (NDCG@10, mean ± std over three seeds). Base denotes the frozen backbone. IHR improves all 30 backbone–domain combinations.

Backbone	Beauty		Toys & Games		Sports & Outdoors		Clothing		Video Games	
	Base	IHR	Base	IHR	Base	IHR	Base	IHR	Base	IHR
GRU4Rec	.0136±.0008	**.0263**±.0004	.0101±.0011	**.0370**±.0017	.0102±.0009	**.0158**±.0006	.0039±.0003	**.0115**±.0006	.0339±.0014	**.0515**±.0010
SASRec	.0271±.0009	**.0367**±.0006	.0365±.0010	**.0542**±.0008	.0127±.0038	**.0191**±.0036	.0093±.0006	**.0165**±.0003	.0462±.0009	**.0604**±.0012
FMLPRec	.0299±.0018	**.0393**±.0003	.0374±.0005	**.0533**±.0008	.0161±.0007	**.0216**±.0004	.0102±.0009	**.0172**±.0005	.0497±.0019	**.0614**±.0016
BERT4Rec	.0381±.0003	**.0443**±.0014	.0375±.0059	**.0530**±.0029	.0213±.0003	**.0249**±.0003	.0105±.0004	**.0178**±.0005	.0643±.0020	**.0743**±.0011
DuoRec	.0577±.0007	**.0607**±.0009	.0650±.0008	**.0723**±.0006	.0304±.0043	**.0336**±.0040	.0198±.0005	**.0234**±.0006	.0789±.0010	**.0859**±.0006
BSARec	.0593±.0008	**.0618**±.0010	.0669±.0007	**.0745**±.0016	.0343±.0002	**.0364**±.0002	.0202±.0003	**.0249**±.0006	.0803±.0006	**.0863**±.0002

Table 3: Comparison with text-enhanced and feature-aware baselines (Recall@10 / NDCG@10, mean ± std over three seeds). All baselines were run from their official repositories with authors' reported hyperparameters under our uniform leave-last-out split and full-ranking protocol. Reported numbers may differ from the original papers due to known protocol-induced shifts in sequential-recommendation metrics [41, 42]. Best in bold, second best underlined.

Method	Beauty		Toys & Games		Sports		Clothing		Video Games	
	R@10	N@10	R@10	N@10	R@10	N@10	R@10	N@10	R@10	N@10
TedRec	.0462±.0008	.0229±.0005	.0394±.0014	.0200±.0006	.0209±.0000	.0103±.0002	.0126±.0001	.0059±.0001	.0685±.0002	.0339±.0005
UniSRec	.0699±.0006	.0328±.0002	.0489±.0017	.0253±.0005	.0125±.0005	.0063±.0002	.0111±.0044	.0055±.0023	.0695±.0015	.0312±.0007
DIF-SR	.0895±.0003	.0438±.0004	.0964±.0008	.0472±.0003	.0534±.0005	.0247±.0002	.0311±.0003	.0142±.0002	.1336±.0006	.0603±.0005
FDSA	.0818±.0006	.0469±.0003	.0837±.0003	.0510±.0002	.0454±.0009	.0254±.0009	.0252±.0006	.0148±.0004	.1216±.0009	.0637±.0003
DuoRec + IHR	.1012±.0009	.0607±.0009	.1176±.0019	.0723±.0006	.0553±.0061	.0336±.0040	.0408±.0009	.0234±.0006	.1548±.0011	.0859±.0006
BSARec + IHR	**.1034**±.0009	**.0618**±.0010	**.1200**±.0021	**.0745**±.0016	**.0612**±.0023	**.0364**±.0002	**.0444**±.0013	**.0249**±.0006	**.1558**±.0003	**.0863**±.0002

Table 4: Hit overlap analysis (Beauty, R@10). "Newly hit" is items newly discovered by IHR. The new hit ratio is (newly hit)/(retained + newly hit), the fraction of combined hits missed by the backbone alone.

Backbone	Retained	Newly hit	Missed	New hit ratio
SASRec	832	683	342	45.1%
BERT4Rec	1,257	546	303	30.3%
BSARec	2,019	292	177	12.6%

Table 5: Profile-source comparison on Beauty. Values are per-seed relative NDCG@10 improvements (%), reported as mean ± std. All methods use the same interpolation framework.

Strategy	SASRec	BERT4Rec	BSARec
RandomRank	−0.6±1.0	0.0±0.0	0.0±0.0
Item2VecRank	+7.2±6.3	+0.3±0.5	0.0±0.0
Doc2VecRank	+25.4±2.6	+9.8±1.2	+2.7±0.8
ReviewEmbRank	+29.0±4.0	+11.6±1.2	+2.6±0.6
IHR (ours)	**+35.3±2.4**	**+16.3±3.9**	**+4.3±0.4**

rather than from our data, suggests a counterintuitive prediction against this: a profile whose construction relies on the same signal source as the backbone (in our setting, item co-occurrence in user sequences) should contribute vanishing improvement regardless of its absolute quality. Item2VecRank is the control that simultaneously satisfies two conditions under this operational definition: it is high-quality (Item2Vec is a widely validated CF embedding) and it is trained on exactly the same item co-occurrence signal as an ID-based sequential backbone. As predicted, Item2VecRank yields +7.2% on SASRec, +0.3% on BERT4Rec, and 0.0% on BSARec, vanishing monotonically as the backbone becomes stronger. Quality is held fixed across the three backbones, so this pattern is incompatible with the quality hypothesis and supports the complementarity hypothesis.

Step 3. The positive side (review-based profiles). Review-based profiles (Doc2Vec, ReviewEmb, TF-IDF) all carry text information the backbone does not use, and all yield positive improvement. The large spread among them (+35.3% vs +25.4% vs +29.0% on SASRec) raises the follow-up question of how a text-based profile should be built, which we address next.

The ablation in Table 6 localizes the effect. IDF is the main driver. Removing it (TF-only) drops performance substantially on every dataset and backbone, for example Beauty SASRec +35.3 → +28.4 and Clothing SASRec +78.0 → +59.5, confirming that how many items a term appears in matters far more than how often it is repeated within reviews. The core signal is therefore the IDF-weighted TF-IDF vector itself, not the downstream SVD compression. Among the dimensionality options, the raw 10,000-dimensional sparse vector is best on most dataset-backbone combinations and Full (with SVD) is best on Video Games, while in nearly every dataset–backbone combination Full is either best or within a few points of the best. We adopt Full not as the strictly highest-scoring setting but as a practical trade-off, since cosine similarity on a 10,000-dimensional sparse vector scales poorly at production catalog sizes while the 500-dimensional compression retains essentially the same quality at a far smaller cost.

Table 6: Ablation across five datasets. Values are per-seed relative NDCG@10 improvements (%), reported as mean ± std. Full uses TF-IDF with SVD and L2 normalization; IDF-only uses binary TF with IDF weighting. Best is bold; second best is underlined.

	Condition	SAS	BERT	BSA
Beauty	Full	35.3±2.4	16.3±3.9	4.3±0.4
	−SVD (raw sparse)	**40.1±3.4**	**18.6±4.1**	**4.9±0.5**
	TF-only (−IDF)	28.4±2.6	11.7±3.3	3.3±0.5
	IDF-only	38.1±3.8	15.7±1.6	4.3±0.2
Toys	Full	48.6±3.4	43.0±16.2	11.5±2.4
	−SVD (raw sparse)	**50.3±4.0**	**44.7±16.4**	**12.4±2.1**
	TF-only (−IDF)	38.2±4.0	34.4±13.9	9.3±1.7
	IDF-only	44.3±3.6	37.1±13.6	8.4±1.2
Sports	Full	55.8±23.1	16.9±1.8	5.9±0.3
	−SVD (raw sparse)	**56.3±19.7**	**17.7±1.6**	5.8±0.6
	TF-only (−IDF)	46.8±21.4	13.9±1.3	4.8±1.0
	IDF-only	51.4±12.2	16.7±1.3	**7.0±1.9**
Clothing	Full	78.0±9.6	69.8±2.0	23.6±1.5
	−SVD (raw sparse)	**85.3±11.0**	**70.9±2.6**	**26.0±1.7**
	TF-only (−IDF)	59.5±9.1	51.1±1.7	18.5±1.6
	IDF-only	64.4±5.7	54.7±4.6	21.7±2.6
VGames	Full	**30.7±1.2**	**15.6±2.3**	**7.5±1.1**
	−SVD (raw sparse)	27.3±0.6	13.8±1.8	7.0±0.9
	TF-only (−IDF)	23.8±0.4	11.7±1.9	5.5±0.5
	IDF-only	21.8±0.7	9.6±1.4	4.6±0.3

5.2.1 Comparison against dense encoders, and the origin of TF-IDF's advantage. To address the concern that a stronger dense encoder might have changed the outcome, we put five general-purpose dense encoders (MiniLM-L12, mpnet-base, Qwen3-Embedding-0.6B and 4B, OpenAI text-embedding-3-large) through the same interpolation framework (3 datasets × 2 backbones × 3 seeds = 18 backbone–dataset–seed evaluations per encoder, 90 across five encoders).

Table 7: Dense-encoder comparison on Toys & Games (mean ± std over three seeds). Δ% is relative improvement over the backbone. z_{mean} measures standardized ground-truth similarity among the backbone's top-100 candidates; Gini measures similarity concentration; Pearson is the correlation between pairwise item similarities induced by the profile and backbone embeddings, with lower absolute values indicating lower representational alignment.

	Profile	NDCG@10	Δ%	z_{mean}	Gini	Pearson
SASRec	TF-IDF	**.0542±.0008**	**+48.5**	**+0.640**	**0.291**	+0.086
	Qwen3-Emb-4B	.0500±.0005	+37.0	+0.437	0.059	+0.059
	Qwen3-Emb-0.6B	.0493±.0004	+35.1	+0.397	0.073	+0.056
	OpenAI-3-large	.0487±.0009	+33.5	+0.306	0.073	+0.028
	mpnet-base	.0443±.0008	+21.6	+0.237	0.111	+0.025
	MiniLM-L12	.0416±.0006	+14.1	+0.202	0.144	+0.019
BSARec	TF-IDF	**.0745±.0016**	**+11.5**	**+0.347**	**0.297**	+0.059
	Qwen3-Emb-4B	.0725±.0006	+8.5	+0.198	0.058	+0.025
	Qwen3-Emb-0.6B	.0723±.0002	+8.1	+0.182	0.071	+0.009
	OpenAI-3-large	.0720±.0003	+7.7	+0.075	0.071	−0.034
	mpnet-base	.0701±.0008	+4.8	+0.062	0.106	−0.019
	MiniLM-L12	.0682±.0006	+1.9	+0.067	0.139	−0.015

Table 7 reports the results together with three diagnostic quantities (z_{mean}, Gini, and Pearson) that characterize each profile. TF-IDF ranks first across every backbone and every domain, and neither Qwen3-Emb-4B nor OpenAI text-embedding-3-large, a top-tier commercial embedding, catches up. This ordering is important for the deployment question in RQ2. OpenAI text-embedding-3-large requires third-party processing of the review text, while Qwen3-Embedding-4B avoids that transfer only by adding local GPU inference. The profile that wins is instead produced by a CPU-only batch job over data the operator already stores. On these benchmarks, keeping review representation local therefore does not impose an observed recommendation-utility penalty. This is an operational processing result, not a claim that local TF-IDF makes the system private. TF-IDF shows a z_{mean} 1.5–3 times higher and a Gini 2–5 times higher than the dense encoders, meaning that it strongly separates the ground truth from non-ground-truth items and concentrates similarity on a small number of candidates, whereas dense encoders spread similarity smoothly across candidates.

Where does this come from? We extract the 20 highest- and 20 lowest-IDF terms per domain (Table 8).

Table 8: Highest- and lowest-IDF terms per domain. High-IDF terms are mostly identifiers (brands, characters, product codes), low-IDF terms are mostly sentiment or context words.

Domain	High-IDF Top (identifier)	Low-IDF Top (semantic-rich)
Beauty	dermanail, schick, pinaud, saccharomyces, foxbrim, kerafuse, swissco, clubman	amazon, feels, happy, scent, quality, clean, highly, reviews
Toys	brobee, drax, digimon, scribblenauts, stratego, cyclonus, zeus, boomco	birthday, played, likes, favorite, happy, best, overall, definitely
Sports	flexfit, biolite, srixon, thermacell, clipdraw, clicgear, madden, noco	set, cheap, heavy, large, worth, perfectly, happy, actually
Clothing	tuckable, 1604, pediped, kangol, moab, ameribag, pugster, daytona	wanted, amazon, shoes, feet, fabric, light, easy, super
Video Games	digimon, chessmaster, rygar, smurf, raziel, codebreaker, beavis, uru	sound, music, levels, control, experience, awesome, world, away

In all five domains, high-IDF terms are often proper nouns, product codes, or domain-specific identifiers (e.g., schick, flexfit, pediped, srixon, digimon, and 1604). These tokens are frequently more item-specific than general semantic descriptors. TF-IDF preserves this lexical specificity by assigning greater weight to terms that occur in fewer item documents. By contrast, lower-IDF terms such as happy, quality, or worth occur across many item documents and are therefore less discriminative at the item level. The diagnostic results in Table 7 are consistent with this interpretation: TF-IDF produces stronger ground-truth separation and a more concentrated similarity distribution than the tested dense encoders. This does not imply that dense semantic representations are generally inferior; rather, in this task, identifier-level specificity appears to provide auxiliary information that the collaborative backbone does not already capture.

5.3 RQ3. Is IHR practical and robust?

We examine robustness to hyperparameters, behavior under top-K reranking, and robustness to review-scarce items, and report the computational cost.

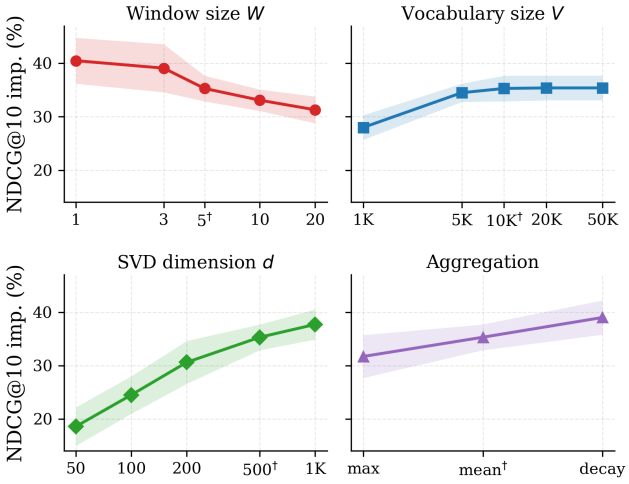


Figure 3: Hyperparameter sensitivity (Beauty, NDCG@10 improvement in %, mean ± std over three seeds) across window size W, SVD dimension d, vocabulary size V, and aggregation for SASRec and BSARec.

5.3.1 Hyperparameter robustness. Window size W is stable for $W \geq 3$, with SASRec peaking at W=1 (+40.5%) and BSARec peaking around W=10 (+4.5%). Since the optimum differs across backbones, we adopt W=5 as a backbone-agnostic default that lies within the stable region for both. SVD dimension d increases monotonically from d=50 to d=500 and improves slightly at d=1,000, so we use d=500 as a tradeoff. Vocabulary size V saturates quickly for $V \geq$ 5,000 (around +34.5 to +35.4% on SASRec), while BSARec is almost invariant across the range. For aggregation, decay gives slightly higher improvement than mean (+39.0 vs +35.3 on SASRec) but introduces an extra hyperparameter, so we default to mean. Max is lowest as it relies on a single extreme item.

5.3.2 Top-K reranking efficiency. Real systems often score only the top-K candidates from a generation stage rather than the full catalog, so we check if IHR remains effective in this two-stage pipeline.

For all three backbones the improvement saturates quickly with K and reaches the full-item level by K=500 or sooner. (Figure 4) Even at K=50, SASRec retains 85% of the full improvement (+29.9 vs +35.3) and BSARec retains 95% (+4.1 vs +4.3). This means that in a two-stage pipeline with a candidate pool of hundreds to a few thousand, IHR realizes essentially all of the gain that would be available from reranking the full catalog, at a cost of $O(K \cdot d)$ which is under 0.01ms at K=1000 and d=500.

5.3.3 Robustness on items with few reviews. We split test items into five groups by review count (Table 9). Improvement is not monotonic in review count. SASRec and BSARec peak at 16–30 reviews (+75.8% and +13.2%, respectively), whereas BERT4Rec peaks at 6–15 reviews (+41.6%). With very few reviews ($\leq$ 5), SASRec and BERT4Rec still improve while BSARec changes by −0.7%. Beyond 60 reviews, gains drop sharply and become slightly negative for BERT4Rec and BSARec. Two effects plausibly compound: popular

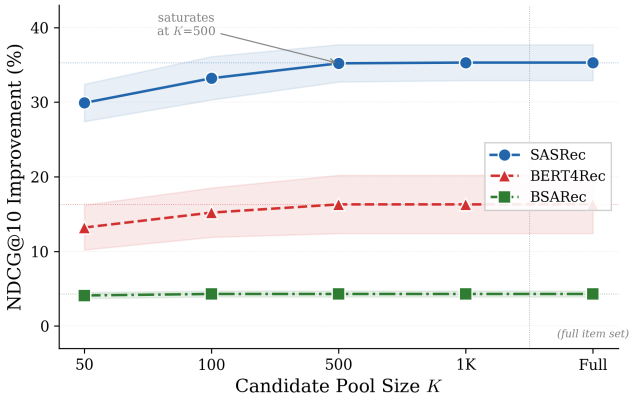


Figure 4: Top-K reranking (Beauty, NDCG@10 improvement in %, mean ± std over three seeds) as a function of the candidate pool size K for SASRec, BERT4Rec, and BSARec. Full denotes reranking the entire item set.

items are already well handled by stronger backbones, and aggregating many reviews introduces more generic terms that can dilute item-specific identifiers. The pattern therefore suggests that review quantity alone is not the driver; profile sharpness and residual room in the backbone both matter. Routing low-sharpness or heavily reviewed items back to the backbone score is a natural direction for future work.

Table 9: NDCG@10 improvement by review count group (Beauty, %).

Review count	n	SASRec	BERT4Rec	BSARec
0–5	4,446	+35.2	+22.7	−0.7
6–15	7,120	+60.2	+41.6	+9.9
16–30	3,794	**+75.8**	**+34.8**	**+13.2**
31–60	3,608	+68.3	+31.3	+9.8
61+	3,395	+10.6	−0.1	−2.3

5.3.4 Computational cost. On Beauty (n_{items}=12,102, n_{users}=22,363) on a single CPU core (Xeon E5-2620 v4 @ 2.10GHz, `taskset -c 0`, no GPU used), text preprocessing takes 2.4s, TF-IDF plus SVD (d=500) and L2 normalization takes 17.3s, and reranking all users with W=5 mean takes 45.9s, for a total of 65.5s and a per-user latency of 2.05ms.

6 Discussion and Limitations

Operational scope. IHR is not a privacy mechanism, and nothing here should be read as a formal guarantee. We make no differential-privacy claim and do not evaluate against an adversary. The operational result is narrower: compared with text-enhanced pipelines that use a hosted encoder and retrain the recommendation model, IHR can keep review representation inside operator-controlled infrastructure and leave the deployed backbone frozen while maintaining or improving utility in the tested settings. The Amazon

datasets used here are public research benchmarks, so these experiments establish utility and deployment properties rather than a measured confidentiality benefit for these particular records.

Auditability and linkability. The TF-IDF profile is inspectable term by term, but the same rare identifiers that contribute to ranking can also facilitate linkage [37]. Item-level aggregation reduces direct correspondence to a single reviewer but is not a privacy guarantee. If profiles are exported, shared across organizations, or reused for transfer, the high-IDF tail should be audited under an explicit threat model rather than assumed safe because construction was local.

Why the classical representation wins. The redundancy–complementarity pattern offers a simple explanation. Dense encoders are optimized for semantic similarity, some of which overlaps with category and co-occurrence structure already captured by a strong ID-based backbone. TF-IDF instead emphasizes item-discriminative identifiers that the backbone cannot observe. The Item2VecRank control supports this interpretation: a profile derived from the same collaborative signal contributes progressively less as the backbone strengthens. The implication is not that sparse representations are universally superior, but that side information should be judged by the new signal it adds to a deployed model.

IHR still has important limitations. It requires reviews to build informative profiles, is evaluated only on English Amazon reviews in e-commerce, and uses item-level aggregation that cannot distinguish users who purchased the same item for different reasons. Performance also degrades for heavily reviewed head items. User-specific profiles, dynamic routing based on profile sharpness, multilingual evaluation, evaluation on private organizational data, and explicit linkage experiments remain for future work.

7 Conclusion

We presented IHR, a post-hoc review reranker for frozen sequential recommenders. Across six backbones and five Amazon domains, IHR improves all 30 evaluated backbone–domain combinations, and the strongest IHR-augmented systems achieve higher ranking metrics than representative retraining-based text-aware systems under the same protocol. Controlled profile comparisons show that locally constructed TF-IDF with SVD outperforms five dense encoders because rare, item-discriminative review terms complement information already captured by ID-based backbones.

The central deployment result is that this gain can be obtained without sending the review corpus to a hosted embedding service or retraining the deployed recommender. This reduces additional third-party processing and deployment dependencies, but it is not a formal privacy guarantee: the same sparse identifiers that improve ranking can create linkability concerns if profiles are disclosed. More broadly, the experiments support a complementarity principle for hybrid AI pipelines: auxiliary representations are most useful when they contribute information the deployed model does not already recover.

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